Project Name -Specifications Document

1. Project overview.

Short description of the DUT.

Explain:

- Purpose of component
- Intended functionality
- The role of this component in a digital system
- General operating principle

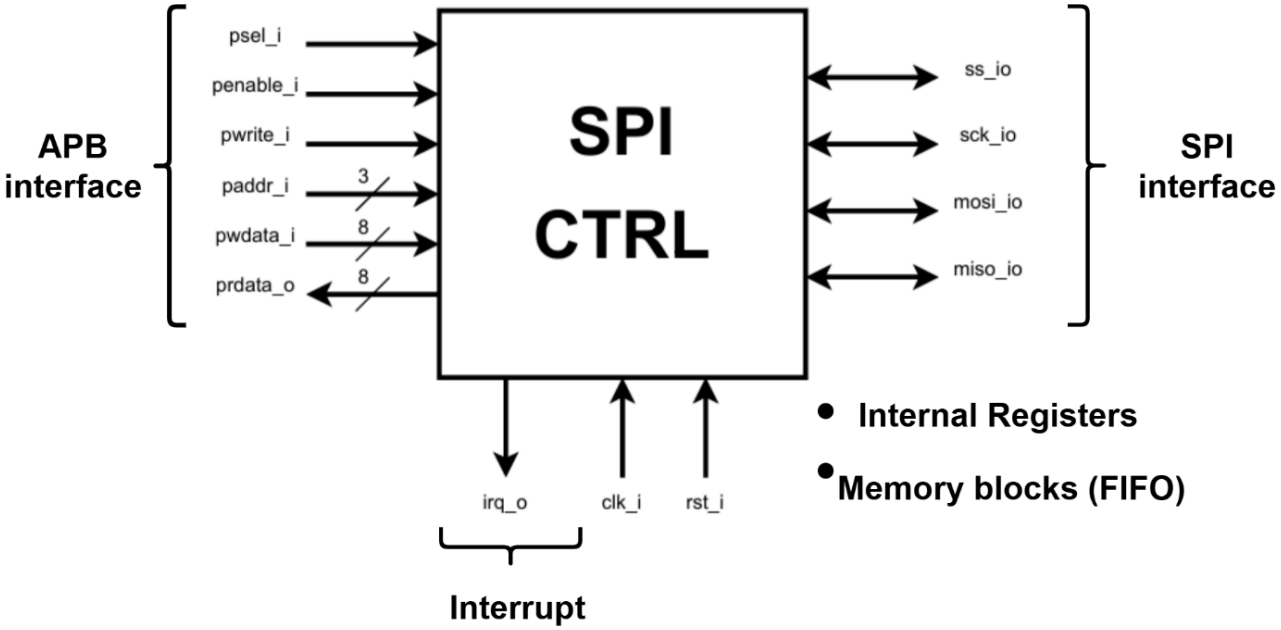


Figure of the DUT.

Define the functionality included in the current implementation.

Specify any functionality that is intentionally excluded.

List relevant refrences and what was used in studying.

2. Interface specification

Decribe all external signals and interfaces of the DUT.

List of the requirements from the specification with each of their identifiers.

Clock and reset signals:

Signal	Direction	Width	Description
clk_i	Input	1	Clock signal
rst_i	input	1	Reset signal

Specify:

- Clock edge used for operation
- Reset polarity
- Syncrhonous or asynchronous reset
- Reset assertion and reset deassertion behavior

Include a table of reset values.

Reg/Signal	Reset value	Description
reg_1	0	
signal_1_i	0	

Input signals:

Signal	Width	Description
input_1_i		
input_1_i		

Describe the purpose and expected behavior of each input.

Output signals:

Signal	Width	Description
output_1_o		
output_2_o		

Describe the purpose and expected behavior of each output.

Interface behavior:

- Signal dependencies
- Handshake requirements
- Data transfer protocol
- Valid operating conditions
- Timing requirements

3. Parameters and configuration.

List of all configurable parameters of the DUT

Parameter	Default value	Description	Constraints
DATA_WIDTH	8	Data bus width	Greater than 0
DEPTH	16	Storage depth	Greater than 0

Explain how changing the parameters affects the DUT.

4. Functional requirements.

List all of the requirements of the DUT.

Each requirements must have a unique identifier and describe one specific, clearly defined behavior.

- REQ-001: The DUT shall clear its internal state when reset is asserted.
- REQ-002: The DUT shall accept input data when write enable signal is asserted.
- REQ-003: The DUT shall maintain its current state when the enable signal is asserted.
- Etc.

5. Operational behavior.

Explain how the DUT operates under different conditions.

a. Normal operation

Sequence of operations and relationships between inputs, internal state and outputs for normal operations.

b. Boundary conditions

Describe the expected behavior at the limits of the DUT's operating range.

- Minimum and maximum counter values
- First and last memory addresses
- Empty and full FIFO conditions
- Minimum and maximum supported data widths

c. Simultaneous operations

Describe the DUT when multiple operations occur at the same time.

- Simultaneous read and write
- Multiple active requrests
- Reset during active operation
- Input updates during ongoing transaction

d. Illegal or unsupported operations

Describe inputs and operation sequences that are outside the supported operating conditions.

Specify the expected DUT behavior, such as:

- Ignoring the operation
- Maintining the previous state
- Returning error indicator
- Entering defined recovery state

e. Error conditions

Describe possible error conditions and how the DUT should respond.

- Invalid addresses
- Invalid commands
- Protocol violations
- Overflow or underflow
- Invalid state transitions

6. Timing specification.

Describe the timing requirements of the DUT.

Include:

- Clocking behavior
- Input sampling conditions
- Output update conditions
- Operation latency
- Minimum and maximum respone times
- Handshake timing
- Data stability requirements

a. Latency and throughput

- Number of clock cycles for an operation
- Whether operations can overlap
- Maximum supported transaction rate
- Wehther b2b transactions are supported

b. Timing diagrams

Include timing diagrams illustrating the expected behavior for important operations.

- Reset sequence
- Read operation
- Write operation
- Req/Ack sequence
- Data transfer

7. Design constraints.

a. Design constraints

- Supported data widths
- Maximum storage depth
- Supported clock frequency
- Supported protocol configurations

8. Specification summary (can be moved to the beginning).

Summarize the main characteristics of the DUT.

Characteristic	**Description**
Component type	
Clock domains	
Reset type	

9. Revision history.

Version	**Date**	**Description of changes**
1.0		Initial specification